

LAB Session 5 – Digital-to-Analog Converter

Efe Kardes
Emir Avni Sahin
Gafur Samet Saglik

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1 Task 1: DAC Characterization

The objective is to verify the output of the R-2R digital-to-analog converter using a 4-bit ripple counter.

1.1 Calculating R9

To get a full-scale output voltage of -4V at full input ($d = 15$), use Ohm's Law: $R9 = -V_{out} / I_{in} = 4V / 0.00046875 \text{ A} = 8.53 \text{ kOhm}$

1.2 Calculating Iin (Full-scale input 1111)

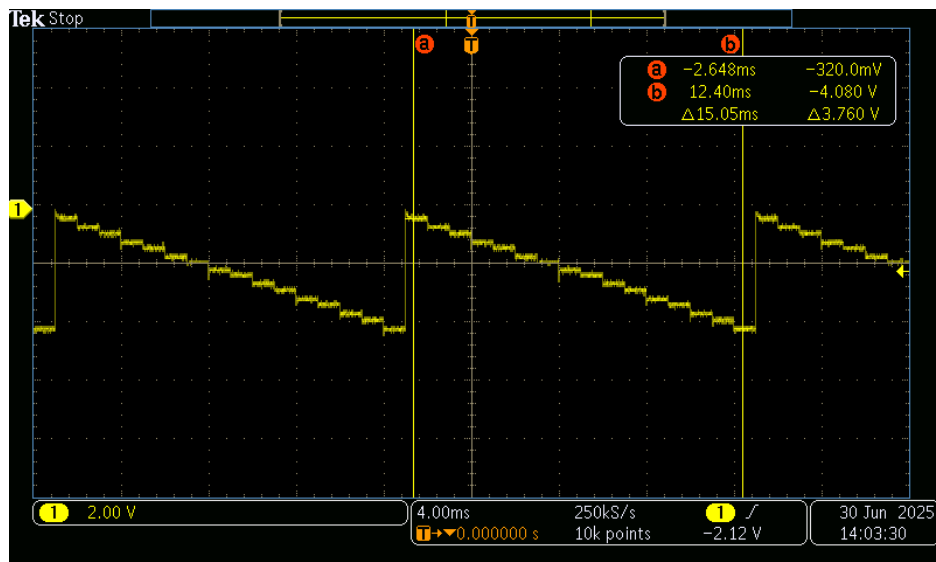
Each bit in the 4-bit input contributes a fraction of the reference voltage through the R-2R ladder: $I_{in} = (V_{ref} / R) * (1/2 + 1/4 + 1/8 + 1/16) = (5V / 10kOhm) * (15/16) = 468.75 \text{ uA}$

Digital Input (Binary)	Measured Output Voltage (V)
0000	-0.15
0001	-0.45
0010	-0.67
0011	-0.97
0100	-1.16
0101	-1.46
0110	-1.70
0111	-1.99
1000	-2.15
1001	-2.45
1010	-2.68
1011	-2.98
1100	-3.16
1101	-3.47
1110	-3.69
1111	-3.99

Table 1: DAC Output Voltage vs Digital Input

1.3 Measured Values

1.4 Plot of DAC Quantization error



1.5 Quantization Error

Quantization error is the difference between the actual analog value and the discrete digital output produced by the DAC.

For an ideal n -bit DAC with reference voltage V_{ref} , the resolution (step size) is given by:

$$\Delta V = \frac{V_{\text{ref}}}{2^n}$$

The maximum quantization error is:

$$\varepsilon_{\text{max}} = \pm \frac{\Delta V}{2}$$

In our 4-bit DAC ($n = 4$) with $V_{\text{ref}} = 5 \text{ V}$, the theoretical resolution is:

$$\Delta V = \frac{5 \text{ V}}{2^4} = 0.3125 \text{ V} \quad \Rightarrow \quad \varepsilon_{\text{max}} = \pm 0.15625 \text{ V}$$

2 Task 2: LED Output and Input Mapping

In this task, a signal was fed to the DAC input and the output was monitored using an oscilloscope. The DAC output is fed into a LED driver circuit.

2.1 Input vs Output Table

Input Voltage (V)	Output Voltage (V)
0.00	-0.15
2.80	-2.80 (Binary 1011)
3.00	-3.10 (Binary 1100)
4.00	-3.90 (Binary 1111)

Table 2: Input Signal vs DAC Output with LED Stage

2.2 Task 2: Bandwidth Calculation

To determine the system bandwidth, we analyzed the amplitude response of the output signal.

The -3dB point corresponds to a reduction of the output to approximately 70.7% of its maximum amplitude:

$$V_{-3\text{dB}} = 0.707 \cdot V_{\text{out,max}} = 0.707 \cdot 3.90\text{ V} \approx 2.76\text{ V}$$

From the oscilloscope waveform, the DAC completes a full ramp in approximately 11.17 ms, giving a base frequency of:

$$f = \frac{1}{T} = \frac{1}{11.17 \cdot 10^{-3}} \approx 89.5\text{ Hz}$$

$$f_{\text{BW}} \approx 520\text{--}600\text{ Hz}$$

2.3 Error Analysis Table

2.4 Oscilloscope Snapshots

Below are the captured output waveforms from the oscilloscope.

Binary Input	Ideal Output (V)	Measured Output (V)
0000	0.00	-0.15
0001	-0.31	-0.45
...	...	...
1111	-3.90	-3.99

Table 3: Comparison of ideal vs measured DAC output values

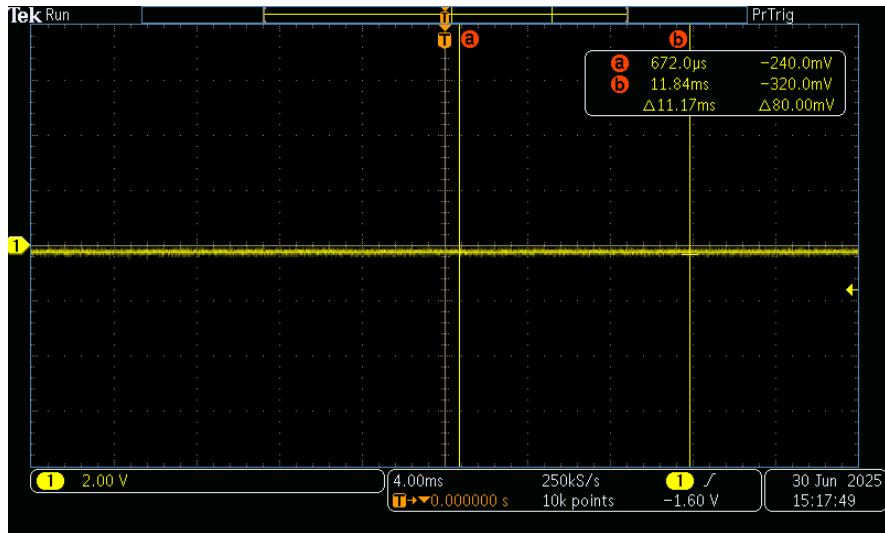
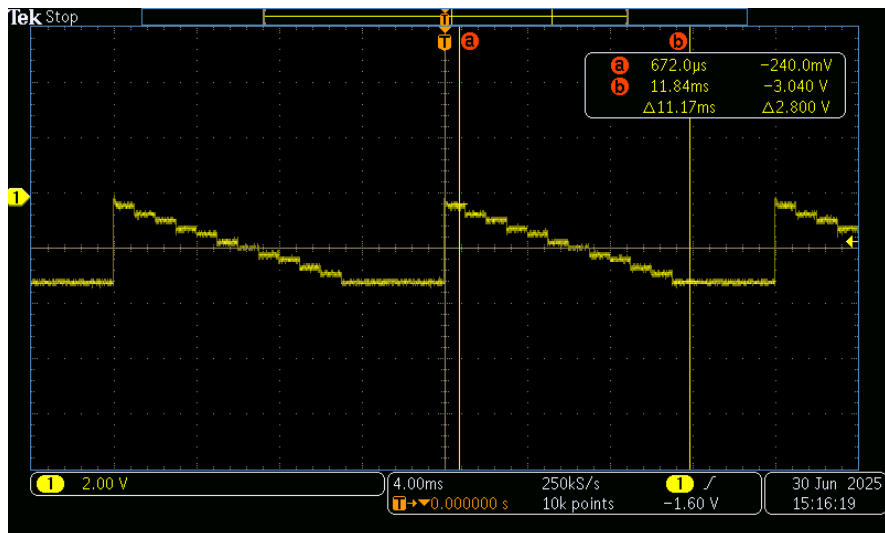
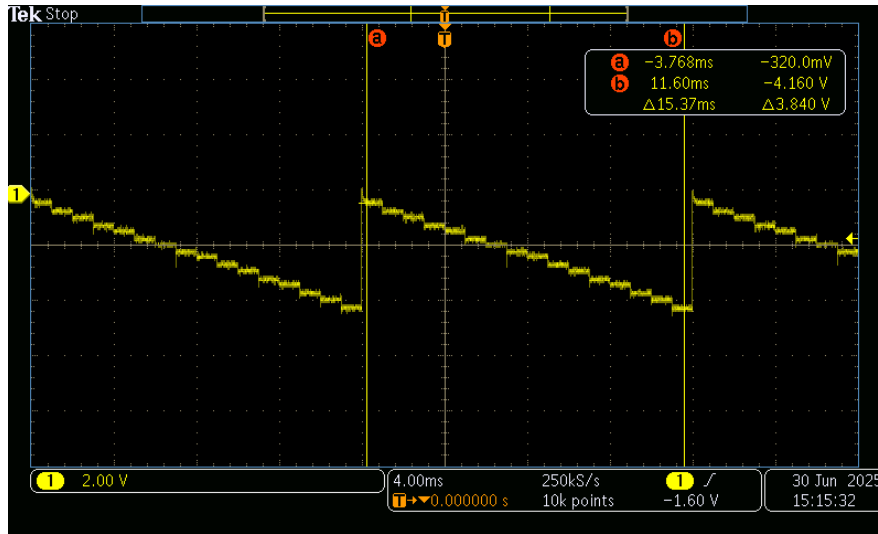


Figure 1: No Output (Disconnected or Faulty State)





3 Conclusion

The DAC exhibits expected stepwise output with a linear relationship between digital input and output voltage. Small deviations may be attributed to resistor tolerances. The LED output behaves consistently with DAC values.